



SVKM's NMIMS
MUKESH PATEL SCH.OF TECH.MGMT ENGG-NM-M

Programme: MASTER OF COMPUTER APPLICATIONS

Year: I/Semester II (Exam Year: 2023-2024)

Subject: Software Engineering

Date: 25 Jun 2024

Time: 10:00 am - 01:00 pm (03:00 Hrs.)

Max Marks: 100

RE EXAMINATION(2023-2024)

Instructions:

1. This question paper contains 2 pages
2. Answer to each new question to be started on a fresh page.
3. Figure in right hand side indicates full marks
4. Question no.1 is compulsory
5. Assume Suitable data if necessary

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| 1) | Answer briefly | 20 |
| | a. Explain the software Engineering code of ethics and professional practice | 5 |
| | b. Describe the concept of a Use Case diagram , give an example of the same. | 5 |
| | c. What are the different types of data modelling in software engineering? | 5 |
| | d. List different types of coupling and describe any two in detail. | 5 |

Answer the following questions(Attempt Any 4 Questions) 80

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| 2) | Answer the following questions | 20 |
| | a. Compare and contrast Incremental and RAD Models. In what scenarios would each be most appropriate? | 10 |
| | b. Describe activity diagram for online shopping application with a neat labelled diagram and explain the components? | 10 |

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| 3) | Answer the following questions | 20 |
| | a. Describe the main components of a DFD and their roles in representing the flow of | 10 |

data within a system also draw the diagram for level 1 DFD in a railway reservation system.

- b. List and explain any one type Software Myths with respect to following software: 10
- a) Zomato
 - b) Amazon Mini TV
 - c) Employees' Provident Fund Organization (EPFO)
- 4) Answer the following questions 20
- a. Describe data flow architectures , give its advantages and disadvantages. 5
 - b. List the golden rules in user interface design? 5
 - c. Determine how waterfall model is appropriate for only some type of system. 10
- 5) Answer the following questions 20
- a. Discuss the principles of visual design and their application in user interface design. 5
 - b. Compare unit testing and integration testing. 5
 - c. Describe class diagram for hospital management system with a neat labelled diagram and explain the components? 10
- 6) Answer the following questions 20
- a. What are common effective software quality assurance activities? 10
 - b. Describe state diagram for student management system with a neat labelled diagram and explain the components? 10
- 7) Answer the following questions 20
- a. Differentiate between data and content coupling. 5
 - b. Explain the intricate relationship between the Software Testing Life Cycle (STLC) and other phases of the Software Development Life Cycle (SDLC). 5
 - c. Explain how the Incremental Model differs from other software development methodologies. 10